

NSSC 2026 | INDIAN INSTITUTE OF TECHNOLOGY, KHARAGPUR

Data Analytics

Unsupervised Anomaly Detection in Mars HiRISE Orbital
Imagery

Executive Summary

This project aims to detect unusual images in 10,422 Mars satellite images without using labels or pretrained models.

- We built an autoencoder to learn and reconstruct normal Martian terrain.
- An Isolation Forest was then used to identify images that were different from the learned patterns.
- A statistical threshold was used instead of an arbitrary cutoff.
- 46 images (0.44%) were identified as anomalous.
- The threshold was 0.526, based on Median + 3×MAD.
- We developed the model through three iterations (v1, v2, v3), fixing issues such as infrastructure limitations, underfitting, and overfitting.
- Finally, we analyzed the limitations of the model and identified areas for future improvement.

In short: The project successfully developed an end-to-end unsupervised pipeline to automatically identify unusual patterns in Mars imagery and explain why those images were flagged.

Introduction & Motivation

The HiRISE camera on NASA's Mars Reconnaissance Orbiter has collected a hundreds of thousands of images of Mars, showing craters, dunes, and other geological features.

Among these images, some may be abnormal, such as:

- Sensor errors
- Corrupted or spliced images
- Non-Martian content

Our goal is to detect these unusual images without using labels or examples of anomalies.

For this, we developed an end-to-end unsupervised pipeline using:

1. A custom **deep autoencoder** to learn and compress normal Mars imagery.
2. An **Isolation Forest** to detect unusual patterns in the learned representation.
3. A **statistical threshold** to determine which images are truly anomalous.
4. **Reconstruction analysis** to understand why an image was flagged.

FACTS !

Think of it like a security guard who has spent years watching employees walk through the same door every day. Nobody ever showed him a photo of an "intruder" — he doesn't need one. He has simply learned the normal pattern so thoroughly that any deviation from it is instantly, intuitively obvious. That is exactly the principle behind this pipeline: we never show the model a single example of what an anomaly looks like. We only teach it what normal Mars terrain looks like, and let deviation from that be the signal

Dataset Overview

The dataset contains 10,422 grayscale images of the Martian surface. Each image is resized to 227×227 pixels. The images come from around 172 unique HiRISE observations.

The dataset is fully unsupervised, meaning no anomaly labels are provided. An unknown number of tampered or foreign images were included at random positions.

Property	Value
Total image crops	10,422
Image dimensions	227 x 227 pixels, grayscale
Unique source observations	~172
Crops per source observation	1 to 278 (median ~ 50)
Metadata fields provided	latitude, longitude, sun_angle, season, resolution
Ground-truth labels	None provided (fully unsupervised task)

An important point is that the number of images from each source is uneven. Some sources provide only a few images, while others provide hundreds. Therefore, during location analysis, we considered the source distribution so that a frequently sampled source was not automatically treated as more anomalous.

Methodology

Our project follows a 9-step pipeline to detect and understand unusual Mars images:

- **Mars Images** – Start with 10,422 unlabeled images.
- **Autoencoder** – A custom CNN learns the normal patterns in Mars images and reconstructs them.
- **Latent Vector** – Each image is converted into a 256-dimensional representation.
- **Isolation Forest** – Detects images that are different from the majority.
- **Novelty Score** – Gives each image a score, where higher = more anomalous.
- **Statistical Threshold** – Uses $\text{Median} + 3 \times \text{MAD} = 0.526$ to identify anomalies instead of choosing an arbitrary number.
- **Anomalous Images** – 46 images (0.44%) were detected as anomalous.
- **Reconstruction & Heatmaps** – The top 5 anomalies are reconstructed and their pixel-level errors are visualized.
- **Physical/Geological Analysis** – Each anomaly is studied to suggest a possible reason for why it was flagged.

FLOW /

Images → Autoencoder → 256D Latent Vector → Isolation Forest → Novelty Score → Statistical Threshold →
Anomalies → Heatmaps → Geological Explanation

PHASE 1 : DEEP LATENT COMPRESSION (AUTOENCODER)

5.1 Architecture

We built a **Convolutional Autoencoder from scratch** to learn the important features of Mars images.

- The **Encoder** gradually reduces the image size from 227×227 to 7×7 using five stride-2 convolution layers.
- A channel-squeeze layer then reduces the channels from 256 to 64.
- The image features are finally converted into a 256-dimensional latent vector.
- The Decoder reverses this process and reconstructs the original 227×227 image.
- The output size was programmatically verified to be exactly 227×227 , without cropping, resizing, or interpolation.
- The complete model has approximately 5.1 million parameters.
- The saved model checkpoint is approximately 61.5 MB, keeping it below the 100 MB GitHub file limit.

BRIEF !

Architecture Flow

227×227 Image $\rightarrow$ Convolution Layers $\rightarrow$ 7×7 Feature Map $\rightarrow$ Channel Squeeze $\rightarrow$ 256D Latent Vector $\rightarrow$ Decoder $\rightarrow$ 227×227 Reconstruction

PHASE 1 : DEEP LATENT COMPRESSION (AUTOENCODER)

5.2 Latent Dimensionality — Design Rationale

We selected a 256-dimensional latent space after considering three main factors:

- Representation: The 256-dimensional bottleneck provides enough capacity to capture important variations in Martian terrain while still forcing the model to learn a compressed representation.
- Model Size: Increasing the latent size to 512 would increase the number of parameters and checkpoint size, making it harder to stay within the 100 MB GitHub limit.
- Anomaly Detection: The latent vector is used by the Isolation Forest. A very small latent space may lose important information, while a very large one can make anomaly detection less effective.

Therefore, 256 dimensions were chosen as a balance between image representation, model size, and anomaly detection performance.

NOTE /**Limitation**

Due to the limited Google Colab GPU time and compute resources, we could not perform a complete experimental comparison between 128, 256, and 512 dimensions. Hence, the choice of 256 was based mainly on architecture and resource considerations, rather than an exhaustive parameter search.

PHASE 1 : DEEP LATENT COMPRESSION (AUTOENCODER)

5.3 Loss Function

To improve the quality of image reconstruction, we used a combined MSE + SSIM loss.

- **MSE** helps the model reduce pixel-level differences.
- **SSIM** helps preserve important image structures such as crater edges, dune patterns, and ridgelines.
- Using only **MSE** produced blurry and overly smooth reconstructions during early experiments.
- SSIM was added to make the reconstructed images more structurally similar to the original images.
- The final value was $\alpha = 0.6$.

$$\text{Loss} = \alpha \times \text{MSE} + (1 - \alpha) \times (1 - \text{SSIM})$$

NOTE !

Why this loss was selected

MSE → accurate pixels but blurry details

SSIM → better structure and texture preservation

MSE + SSIM → better overall reconstruction

PHASE 1 : DEEP LATENT COMPRESSION (AUTOENCODER)

5.4 Training, Regularization, and Early Stopping

Parameter	Value
Optimizer	Adam
Learning Rate	1×10^{-3}
Weight Decay	1×10^{-5}
Dropout	0.3
Early Stopping Patience	15 epochs
Training Duration	~90 epochs

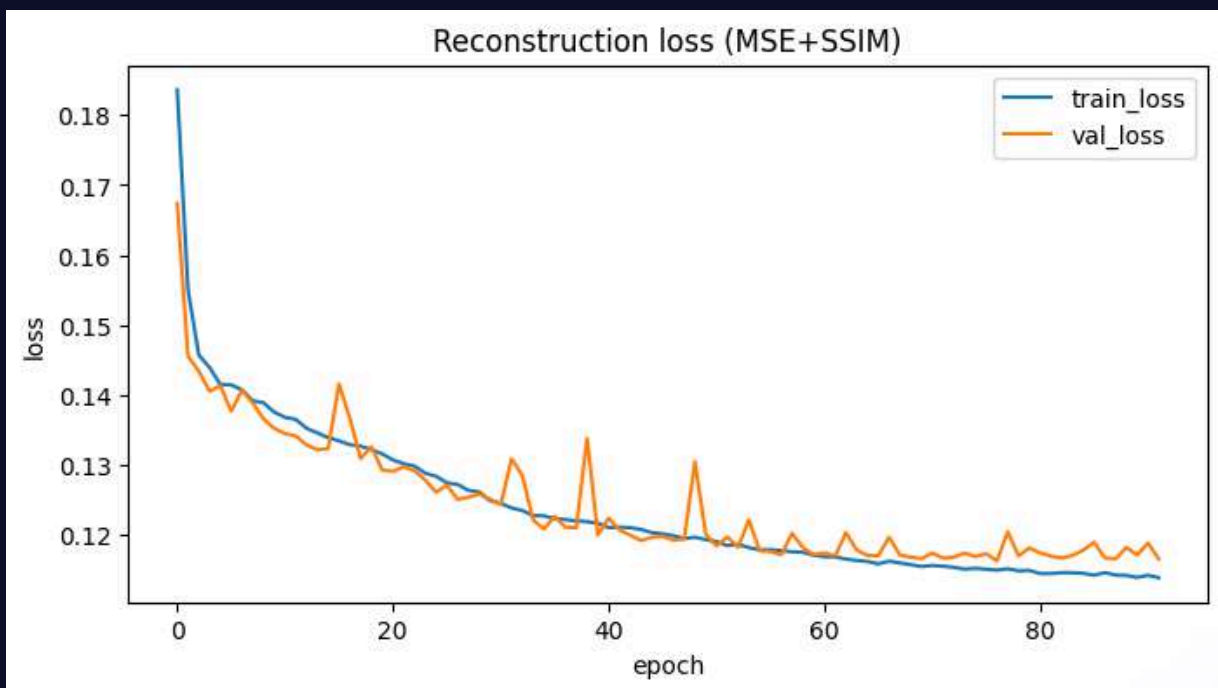
NOTE !**Key Points**

- **Dropout (0.3)** was applied at the bottleneck to **reduce overfitting**.
- **Early stopping** monitored **validation loss** and stopped training when the model **stopped improving for 15 epochs**.
- Training automatically stopped at around epoch 90, instead of using a fixed number of epochs.
- The best checkpoint, based on the **lowest validation loss**, was selected for all further anomaly detection and heatmap analysis.

PHASE 1 : DEEP LATENT COMPRESSION (AUTOENCODER)

5.4.1 Training and Validation Loss

The two loss curves remain close to each other throughout training, indicating that the **model was learning consistently without significant overfitting**. The use of dropout and weight decay helped control the overfitting observed in an earlier model version.

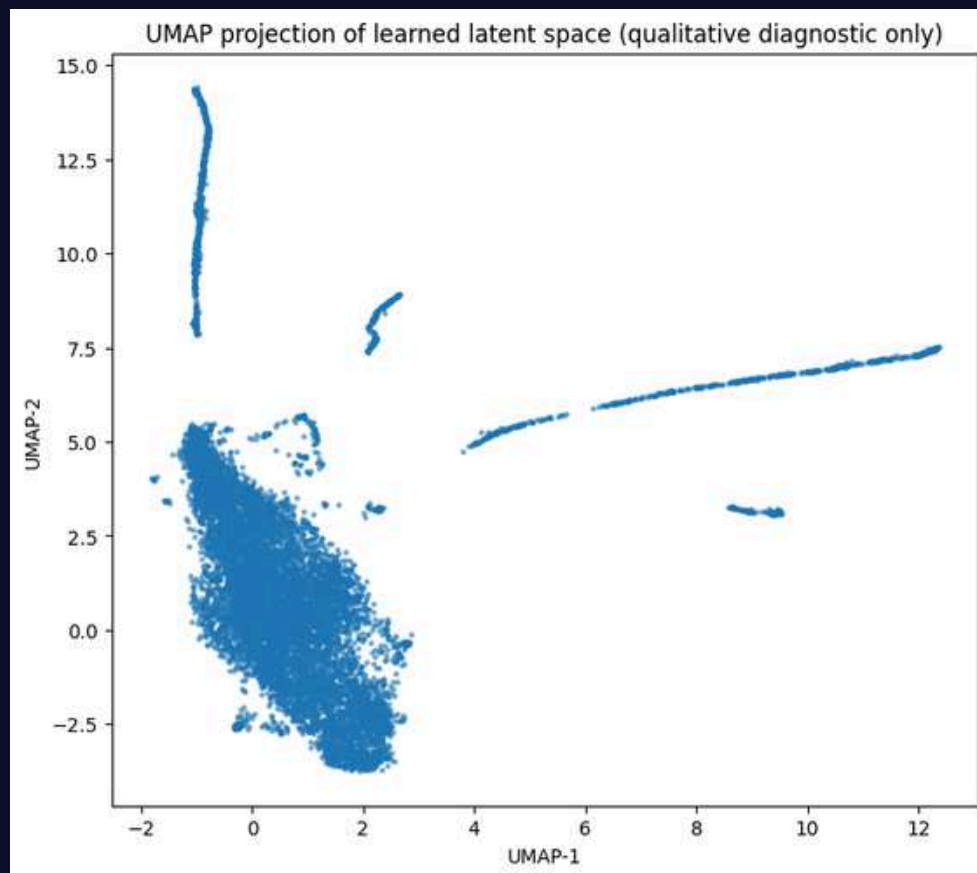
**NOTE /**

Key takeaway : The close training and validation losses indicate that the final model generalizes better than the earlier iteration.

5.5 Latent Space Visualization

We used UMAP to visualize the 256-dimensional latent vectors of all 10,422 images in 2D.

- UMAP was used only as a visual diagnostic tool.
- The visualization showed one large dense cluster with several thin “arm” structures branching away from it.
- One of these arms contained images that were also flagged as anomalous by the Isolation Forest, showing agreement between the two methods.
- Another separate arm contained a legitimate but visually different terrain type, showing that being unusual-looking does not necessarily mean being anomalous.



NOTE !

Key takeaway : UMAP helped us understand the structure of the latent space and provided additional evidence for some anomalies, but it was not used as the final anomaly detector.

PHASE 2 : ISOLATION FOREST NOVELTY ENGINE

6.1 Novelty Scoring

The 256-dimensional latent vectors generated by the trained autoencoder were given to an Isolation Forest for anomaly detection.

Parameter	Value
Input	256D latent vectors
Algorithm	Isolation Forest
Number of Estimators	300
Contamination	auto

The Isolation Forest works on the latent representation instead of raw pixels, making the detection process more efficient while using the features learned by the autoencoder.

Scikit-learn's `score_samples()` gives lower scores for more abnormal images, which is opposite to the competition's required convention.

Novelty Score = – Isolation Forest Score

NOTE !

Key Takeaway

256D Latent Vector → Isolation Forest → Novelty Score → Higher Score = Higher Anomalousness

Higher Novelty Score = More Anomalous

6.2 Statistical Thresholding

Instead of choosing an arbitrary number of anomalies, we **calculated the threshold directly from the Novelty Score distribution**.

First, a normality test showed that the scores were not **normally distributed and were right-skewed**. Therefore, using a **simple mean + k × standard deviation approach** would not be reliable.

Parameter	Value
Mean Novelty Score	0.4202
Median Novelty Score	0.4166
Standard Deviation	0.0343
75th Percentile	0.4433
Maximum Score	0.5806
Normality Test p-value	< 0.000001

Because the distribution is skewed, we used the more robust Median Absolute Deviation (MAD) method.
Higher Novelty Score = More Anomalous

NOTE !

$$\text{Threshold} = \text{Median} + 3 \times \text{MAD}$$
$$\rightarrow = 0.4166 + 3 (0.0363) = 0.5256$$

- Any image with a **Novelty Score > 0.5256** was classified as statistically anomalous.
- Using this threshold, **46 out of 10,422 images (0.44%)** were identified as anomalous.

PHASE 2 : ISOLATION FOREST NOVELTY ENGINE

6.2.1 Threshold Validation

The selected threshold was also visually checked using the **sorted Novelty Score plot and histogram**.

The graphs **showed a clear elbow/separation in the high-score region**, supporting the statistically calculated **threshold of 0.5256**.

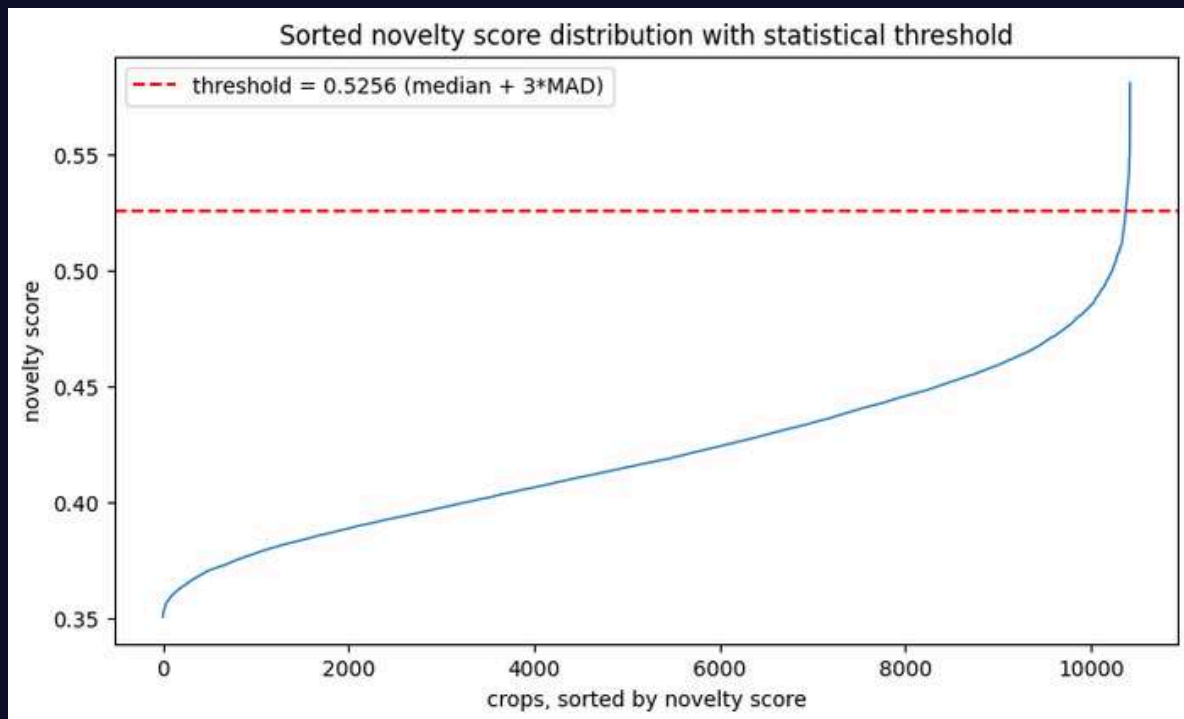


Figure 3 shows the Novelty Scores of all 10,422 images, sorted from lowest to highest.

- Most images have scores in the normal range.
- Near the upper end, the **curve shows a sharp rise (elbow)**.
- The **red dashed line represents the statistical threshold of 0.5256**.
- The elbow occurs close to this threshold, **indicating that the highest-scoring images form a separate outlier region**.

NOTE !

Key takeaway : The visual elbow and the statistically calculated threshold agree, providing additional support for the selected anomaly boundary.

6.2.2 Novelty Score Histogram

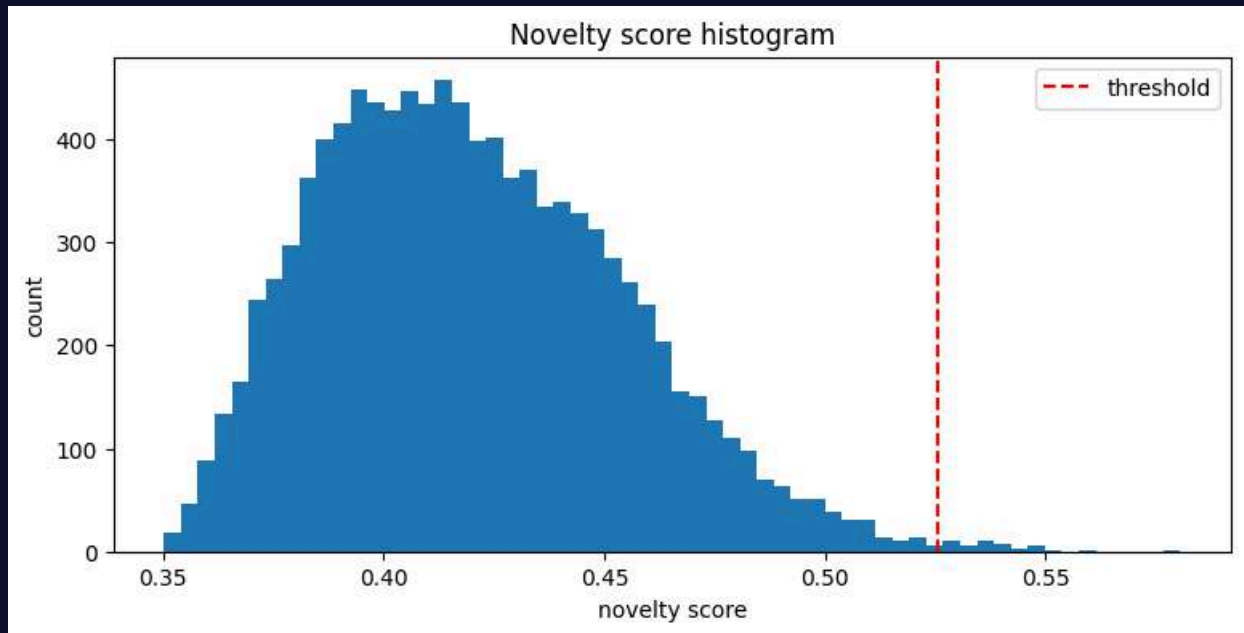


Figure 4 shows the distribution of Novelty Scores for all 10,422 images.

- Most images have scores between approximately 0.35 and 0.50.
- The distribution is right-skewed, with a small number of images having much higher scores.
- The red dashed line shows the selected threshold of 0.5256.
- Only a small number of images lie beyond this threshold, representing the detected anomalies.

Using the selected statistical threshold:

- 46 out of 10,422 images (0.44%) were classified as anomalous.
- The threshold of 0.5256 is 0.0823 above the 75th percentile (0.4433).
- This clear separation indicates that the flagged images belong to the high-score tail rather than being an arbitrary selection from the main distribution.

NOTE !

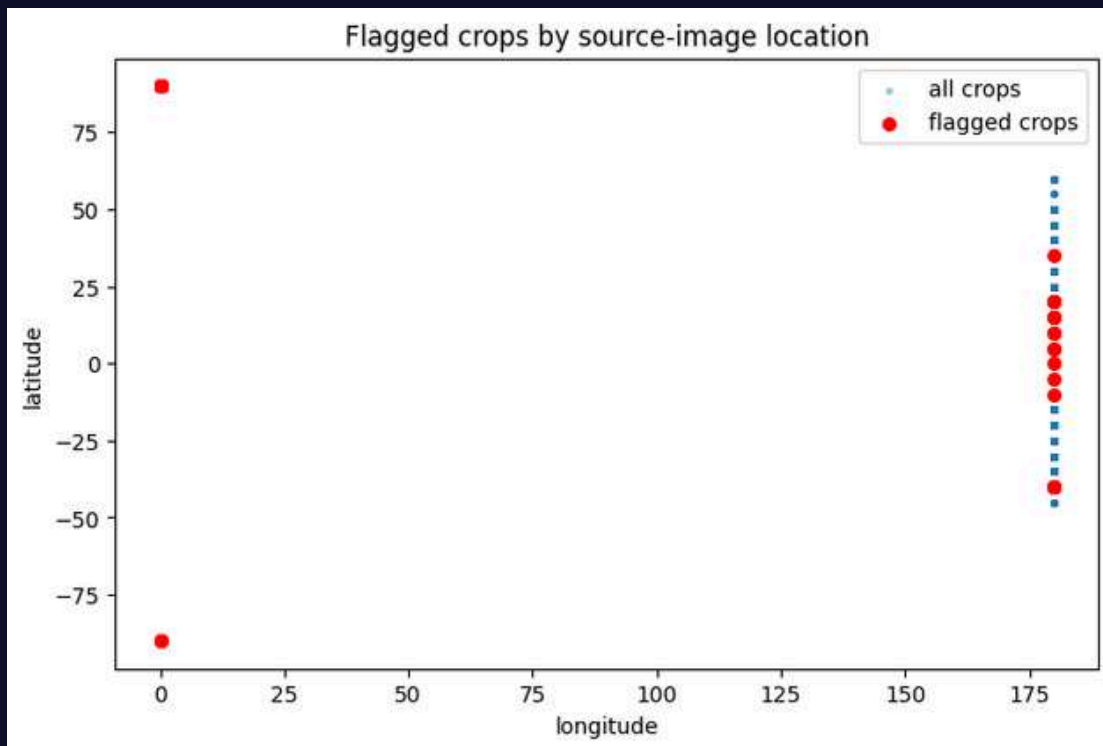
Key takeaway : Only a small portion of the dataset falls into the anomalous region, supporting the effectiveness of the statistically derived threshold.

PHASE 2 : ISOLATION FOREST NOVELTY ENGINE

6.3 Image Location Analysis

We linked the flagged images back to their original HiRISE source observations using the provided metadata files.

- The metadata join was **completed successfully with no missing values**.
- We **analyzed latitude, longitude, sun angle, season, and resolution**.
- No external metadata scraping was used.



The figure shows **flagged crops in red** and **all other crops in light blue**.

- Most flagged images are **concentrated around one longitude band**.
- They cover a **wide range of latitudes, approximately -40° to $+60^{\circ}$** .
- Two flagged groups appear near the opposite longitude, close to the northern and southern extremes.
- This pattern reflects the structure of the provided source metadata and should not be interpreted as proof of a real geological concentration of anomalies.

6.3.I Source Observation Analysis

Source ID	Flagged Images
SRC_039	6
SRC_108	5
SRC_064	5
SRC_044	3
SRC_091	3
SRC_166	3
SRC_128	3
SRC_119	3
SRC_022	2
SRC_071	2

NOTE /

Key takeaway : The detected **anomalies** are distributed across multiple source observations **rather than being concentrated** in one particular observation.

PHASE 2 : ISOLATION FOREST NOVELTY ENGINE

6.4 Optional: Metadata Fusion

As an additional experiment, we combined the 256-dimensional image features with acquisition metadata.

- Sun angle was **normalized**.
- Season was **one-hot encoded**.
- These features were added to the image's latent vector.
- The combined representation was then given to a new Isolation Forest.

The results showed a **large overlap with the original image-only anomaly results**, although some images were newly flagged and some previously flagged images were no longer flagged.

This suggests that metadata provides some additional anomaly information, but the image features remain the main source of novelty information.

NOTE !

Key Takeaway

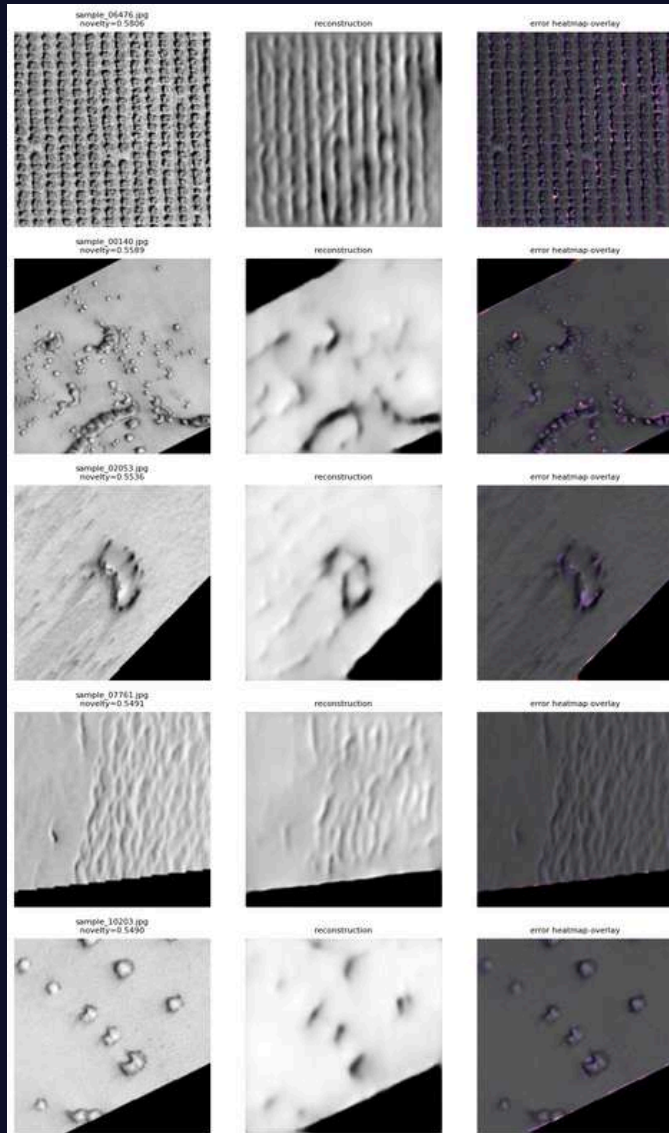
Image Features + Metadata → Slightly different anomaly set

7.1 Pixel-Wise Error Heatmaps

After applying the statistical threshold, we selected the five images with the highest Novelty Scores from the flagged set.

Each selected image was passed through the trained decoder to generate its reconstruction. We then calculated the pixel-wise squared error between the original image and its reconstruction.

The error was visualized as a heatmap, allowing us to identify the regions where the autoencoder had the most difficulty reconstructing the image.



The figure shows, from left to right:

- Original image
- Decoder reconstruction
- Pixel-wise squared-error heatmap
-

The heatmaps highlight the specific regions responsible for the high reconstruction error, making the anomaly detection process more interpretable.

NOTE !

Key takeaway : Instead of only identifying an image as anomalous, the reconstruction heatmaps show where the model struggled and which parts of the image contributed most to its anomalous score.

7.2 Geological Hypotheses

For each of the five most anomalous images, we analyzed the heatmap pattern and developed a possible explanation for why the model flagged it. These are hypotheses, not confirmed classifications.

Crop	Novelty	Observed Pattern	Likely Category
sample_06476.jpg	0.581	Diffuse error across a regular ripple pattern	Genesis Outlier at all
sample_00140.jpg	0.559	Error around multiple smooth, rounded blobs	Domain-shifted terrestrial content
sample_02053.jpg	0.554	Error concentrated around a single closed-loop shape	Semantic splicing
sample_07761.jpg	0.549	Error follows a vertical texture band	Synthetic sensor artifact (tentative)
sample_10203.jpg	0.549	Error concentrated around several rounded blobs	Domain-shifted terrestrial content

NOTE !

The heatmaps show that the model's errors are concentrated around specific structures rather than being completely random.

Interestingly, sample_07761.jpg and sample_10203.jpg show related unusual patterns, providing some consistency between independently flagged images.

Key takeaway : The heatmaps help explain why these images were considered anomalous, while the geological interpretations remain plausible hypotheses rather than definitive classifications.

PHASE 4 : ARCHITECTURE ITERATION & DESIGN JOURNAL

v1 → v2: Checkpoint Size Failure

Symptom:

- The model checkpoints could not be uploaded to GitHub because the files exceeded the 100 MB per-file limit.

Diagnosis:

- The original model flattened a large 256-channel × 14×14 feature map, creating 50,176 values before the fully connected layers. This resulted in approximately 26 million parameters.

Fix:

- An additional stride-2 downsampling layer (14 → 7)
- A 1×1 channel-squeeze layer (256 → 32)

This reduced the flatten size from 50,176 to 1,568 values and significantly reduced the model size.

Outcome:

The total model size decreased to approximately 4.3 million parameters, and the checkpoint size dropped to around 52 MB, allowing successful GitHub uploads and continued training across Colab sessions.

NOTE !**Key Takeaway**

v1: Model too large → v2: Reduced feature-map size → Checkpoint successfully reduced below 100 MB.

v2 → v3: Underfitting (Generic Blurred Reconstructions)

Symptom:

- The **model produced almost the same blurry reconstruction for every image**, regardless of the input. This also caused some simple foreign-content images to receive falsely low novelty scores because they were close to the generic blur.

Diagnosis:

- The 32-channel bottleneck introduced in v2 compressed the information too much. **The decoder did not receive enough information to reconstruct important image structures.**

Fix:

- The **channel squeeze was increased from 32 to 64 channels.**
- Flatten dimension: **1,568 → 3,136**
- Parameters: **~4.3M → ~5.1M**
- Checkpoint size: **~52 MB → ~61.5 MB**

The model still remained below the 100 MB limit.

Outcome:

The reconstructions became more input-specific, preserving features such as ripples, streaks, and dark structures instead of producing a generic, content-independent blur.

NOTE !

Key Takeaway

v2: Too much compression → Generic blurry output

v3: Increased bottleneck capacity → Better structural and image-specific reconstruction.

PHASE 4 : ARCHITECTURE ITERATION & DESIGN JOURNAL

v3 → v4: Overfitting (Validation Loss Divergence)

Symptom:

- After the v3 improvement, the validation loss started increasing around epoch 12, while the training loss continued to decrease. This showed that the model was beginning to overfit the training data.

Diagnosis:

- The wider 64-channel bottleneck increased the model's capacity. Without regularization, the model started learning training-specific details instead of general patterns.

Fix:

- Dropout = 0.3 at the bottleneck
- Weight decay = 1×10^{-5} with Adam
- Early stopping = 15 epochs patience based on validation loss

Outcome:

As shown in Figure 1, the training and validation losses remained much closer together. Training automatically stopped around epoch 90, avoiding unnecessary further training.

NOTE !**Key Takeaway**

v3: Higher capacity → Overfitting

v4: Dropout + weight decay + early stopping → Better generalization and controlled overfitting.

Known Limitation: Contrast-Dependent Recall

To understand the limitations of our final pipeline, we **manually tested it on four candidate foreign-content images** identified during early dataset inspection.

Crop	Pixel Std. Dev.	Flagged?
sample_00140.jpg (high-contrast)	0.228	Yes
sample_10203.jpg (high-contrast)	0.289	Yes
sample_02804.jpg (low-contrast)	0.062	No
sample_02898.jpg (low-contrast)	0.052	No

Observation

- The **pipeline successfully detected the high-contrast foreign-content images but missed the low-contrast versions of the same type.**
- **This shows that the anomaly score is influenced not only by the content of the image, but also by its contrast level.**

Key Limitation

- **High-contrast anomalies → Detected**
- **Low-contrast anomalies → Often missed - Less Chance**

NOTE !

Key takeaway : The final pipeline performs well for visually prominent anomalies but has **reduced recall for low-contrast anomalies**. This limitation was consistently observed across the tested model versions.

Results Summary & Key Strengths

- 1) **Statistical Approach** : The anomaly threshold was selected using the D'Agostino normality test and the **robust Median + MAD method** instead of an arbitrary number of images.
- 2) **Agreement Between Methods** : The **UMAP visualization and Isolation Forest produced overlapping observations** for some anomalous images, providing additional confidence in the results.
- 3) **Iterative Engineering** : The **model was improved through four versions (v1–v4)**. Each change was based on an observed problem and supported by evidence such as parameter count, checkpoint size, and loss behavior.
- 4) **Honest Limitation Analysis** : We tested the **final model for weaknesses and identified a contrast-dependent detection limitation**, particularly for low-contrast anomalies.
- 5) **Competition Compliance** : The complete pipeline was **developed without pretrained weights**, pretrained feature extractors, or transfer learning, following the competition requirements.

NOTE !

Overall Strength

The project combines statistical rigor, unsupervised learning, iterative engineering, interpretability, and transparent analysis of limitations.

Limitations & Future Work

1) **Latent Dimension** : We selected 256 dimensions, but due to limited time and computing resources, we could not fully compare 128, 256, and 512 dimensions.

→ **Future Work** : Train all three versions and compare their reconstruction quality and Isolation Forest performance.

2) **Contrast Sensitivity** : The model performs better on high-contrast anomalies but can miss low-contrast ones.

→ **Future Work** : Add a normalized contrast or texture-variance feature along with the latent vector to reduce this limitation.

3) **Lack of Ground - Truth Labels** : The dataset does not provide confirmed anomaly labels, so traditional precision and recall cannot be directly calculated.

→ **Future Work** : Manually inspect a small sample, such as 40 flagged and unflagged crops, and compare the model's results with human judgment.

NOTE !

Key Takeaway

The current system provides a strong unsupervised solution, but better latent-dimension testing, improved contrast handling, and human-verified evaluation could further improve its reliability.

Tech Stack

Model Architecture

- Framework: PyTorch
- Model type: Custom convolutional autoencoder (encoder-decoder), built entirely from scratch — no pretrained weights, no transfer learning
- Layers used: nn.Conv2d, nn.ConvTranspose2d, nn.BatchNorm2d, nn.LeakyReLU, nn.Dropout, nn.Linear, nn.Sigmoid

Model Specs

- Input/output: 227×227 grayscale images
- Bottleneck path: 227→113→56→28→14→7 (encoder), mirrored in reverse for the decoder
- Latent dimension: 256
- Total parameters: ~5.1 million

Loss Function

- Combined MSE + SSIM
- Mixing weight: alpha = 0.6

Optimizer & Regularization

- Optimizer: Adam (lr = 1e-3, weight_decay = 1e-5)
- Regularization: Dropout (p = 0.3) at the bottleneck
- Training control: Early stopping (patience = 15 epochs on validation loss)

Novelty Detection

- Algorithm: Isolation Forest (scikit-learn)
- Estimators: 300
- Input: the 256-dim latent vectors from the autoencoder (not raw pixels)

Environment:

- Google Colab (T4 GPU)
- Google Drive (storage)
- GitHub (checkpointing)

Conclusion

This project demonstrated that it is possible to detect unusual Mars images without using anomaly labels or pretrained models.

A custom convolutional autoencoder learned the normal patterns of Martian terrain and converted each image into a compact latent representation. An Isolation Forest then used these representations to identify unusual images, while a statistical threshold separated anomalous images from the normal data. Reconstruction-error heatmaps further helped explain which parts of the images caused the model to flag them.

The model was improved through four iterations, addressing checkpoint-size limitations, underfitting, and overfitting. We also identified an important limitation: the system is more effective for high-contrast anomalies than low-contrast ones.

Overall, the project provides a complete, interpretable, and statistically justified unsupervised anomaly-detection pipeline, while clearly documenting both its strengths and limitations.



THE END

Thank you For Your Time and Attention

Our pipeline successfully detected 46 anomalous images (0.44%) from 10,422 Mars images using fully unsupervised learning. The autoencoder, Isolation Forest, statistical thresholding, and heatmaps provided an interpretable end-to-end solution.

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